UNIVERSITY OF UYO
FACULTY OF PHYSICAL SCIENCES
DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION SESSION:2025/2026
COURSE CODE: MTH 214(VECTOR ANALYSIS)
TIME: 2 HOURS
INSTRUCTION: ANSWER FOUR(4) QUESTIONS ONLY

1(a) If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, show that $A = \sqrt{\mathbf{A}\cdot\mathbf{A}} = \sqrt{A_1^2 + A_2^2 + A_3^2}$

(b) If $\mathbf{A} = \mathbf{i} + \mathbf{j}$, $\mathbf{B} = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$, and $\mathbf{C} = 4\mathbf{j} - 3\mathbf{k}$, find

(i) $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$

(ii) $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$

2(a) If $\mathbf{r} = (t^3 + 2t)\mathbf{i} - 3e^{-2t}\mathbf{j} + 2\sin 5t\mathbf{k}$, find

(i) $\left|\dfrac{d\mathbf{r}}{dt}\right|$ at $t = 0$

(ii) $\left|\dfrac{d^2\mathbf{r}}{dt^2}\right|$ at $t = 0$

(b) Prove that $\dfrac{d}{du}(\mathbf{A}\cdot\mathbf{B}) = A\cdot\dfrac{d\mathbf{B}}{du} + \dfrac{d\mathbf{A}}{du}\cdot\mathbf{B}$ where $\mathbf{A}$ and $\mathbf{B}$ are differentiable functions of u.

3(a) If $\phi(x, y, z) = x^2yz$ and $\mathbf{A} = 3x^2y\mathbf{i} + yz^2\mathbf{j} - xz\mathbf{k}$, find $\dfrac{\partial^2}{\partial y\partial z}(\phi\mathbf{A})$ at the point $(1, -2, -1)$.

(b) If $\mathbf{A} = x^2\sin y\mathbf{i} + z^2\cos y\mathbf{j} - xy^2\mathbf{k}$, find $d\mathbf{A}$.

4(a) If $\phi = x^2yz^3$ and $\mathbf{A} = xz\mathbf{i} - y^2\mathbf{j} + 2x^2y\mathbf{k}$, find

(i) $\nabla \times \mathbf{A}$

(ii) $\text{curl}(\phi\mathbf{A})$

(b) Prove $\nabla\cdot(\phi\mathbf{A}) = (\nabla\phi)\cdot\mathbf{A} + \phi(\nabla\cdot\mathbf{A})$

$\frac{dr}{dt}$

5(a) If $\phi = 2x^2y - xz^3$, find

(i) $\nabla\phi$ and

(ii) $\nabla^2\phi$

(b) Prove div curl $\mathbf{A} = 0$.